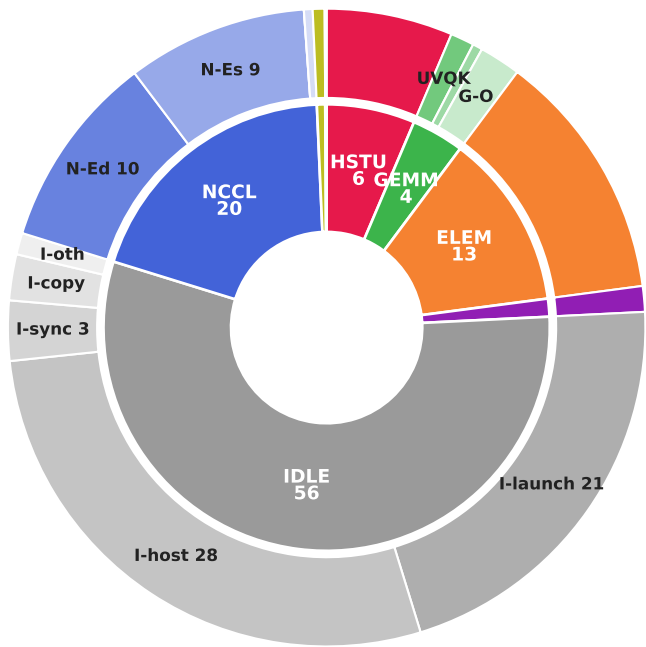
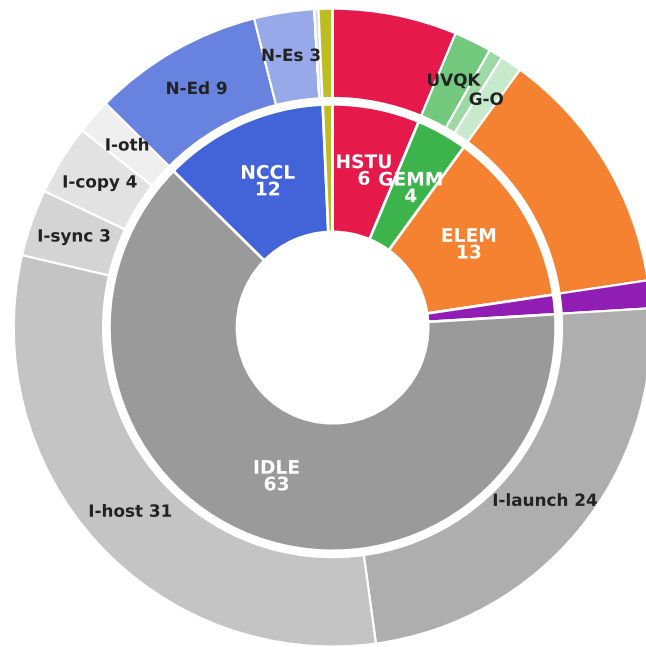


§8 scale-study GPU-time sunburst — GB300 16-GPU, exposed accounting, time-weighted aggregate over 80 steps
inner ring = grouping · outer ring = leaves · % of the step (exp4 = §5.2(B) fastest-step ref)

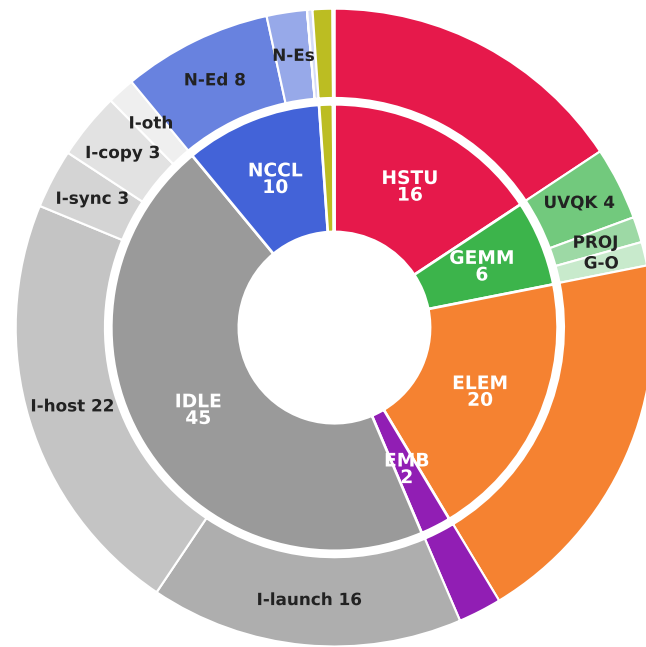
exp4 (ref, fastest) · 50M/2048



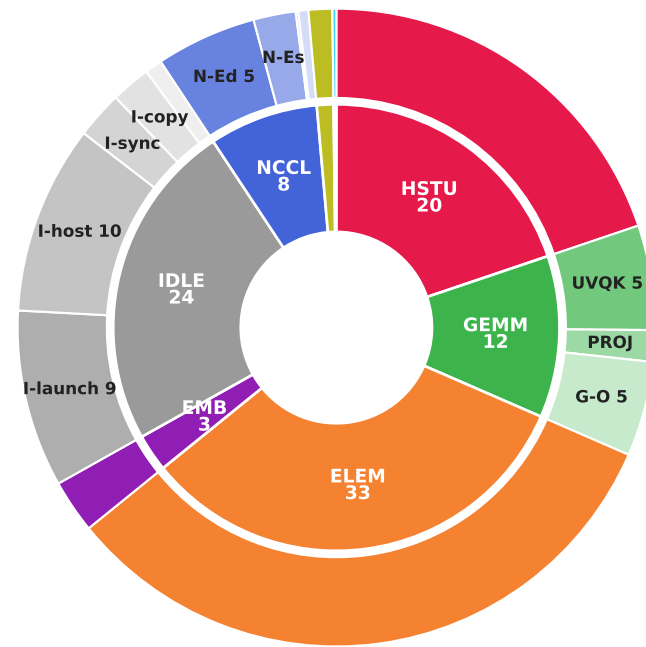
exp5 · 50M/2048 (80-step agg)



zipf · 1B/4096 (80-step agg)



lognormal · 1B/4096 (80-step agg)



- | | | | | | |
|----------------------------|-----------------------|---|---|----------------------------|--------------------|
| ■ HSTU = hstu fwd/bwd | ■ G-O = gemm / others | ■ N-Ed = nccl DENSE/all-reduce (exposed) | ■ I-launch = idle: kernel-launch (CPU dispatch) | ■ I-sync = idle: sync wait | ■ OTH = others |
| ■ UVQK = gemm / uvqk | ■ ELEM = elementwise | ■ N-Es = nccl SPARSE/all-to-all (exposed) | ■ I-host = idle: host/python gap | ■ I-copy = idle: H↔D copy | ■ OVL = overlapped |
| ■ PROJ = gemm / projection | ■ EMB = embedding op | ■ N-Os = nccl sparse (overlap) | | | |